

UNIT 2: Design Thinking & Human-Centred AI



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- Explainable AI (conceptual overview)

Introduction to Design Thinking



- Design Thinking in Human-Centered AI (HCAI) integrates empathetic, user-focused methodologies with AI capability to create ethical, transparent, and collaborative technologies.
- It shifts focus from mere automation to augmenting human intelligence, ensuring systems are trustworthy, fair, and designed to address genuine human needs rather than just technical possibilities

Key aspects of Design Thinking in HCAI



- **Empathy & User-Centricity**
- **AI as a Partner**
- **Iterative Process**
- **Collaboration**

Why Design Thinking is Important



- Focuses on **user needs**
- Encourages **creative solutions**
- Reduces risk of product failure
- Improves **user experience**
- Helps build **better AI products**

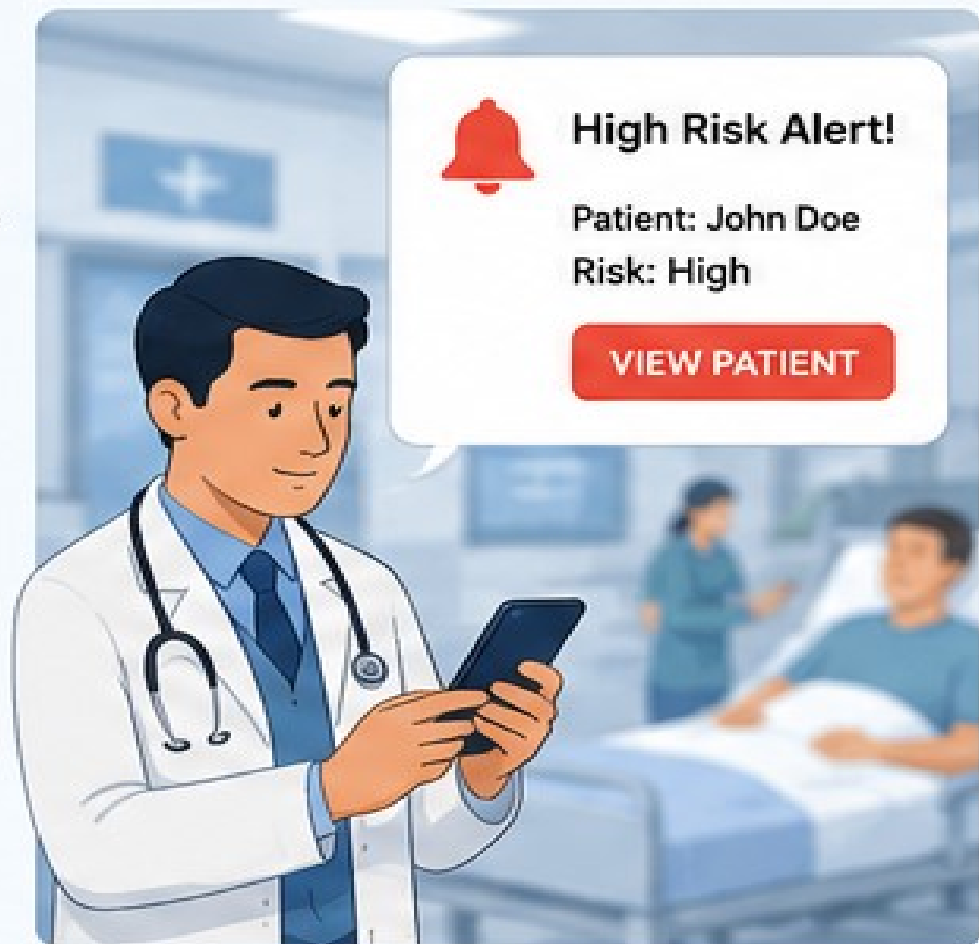
1 Focuses on User Needs

Example:

A hospital wants to use AI for patient care.

Doctors need quick alerts for high-risk patients instead of more data.

AI sends simple notifications, saving time and improving patient care.

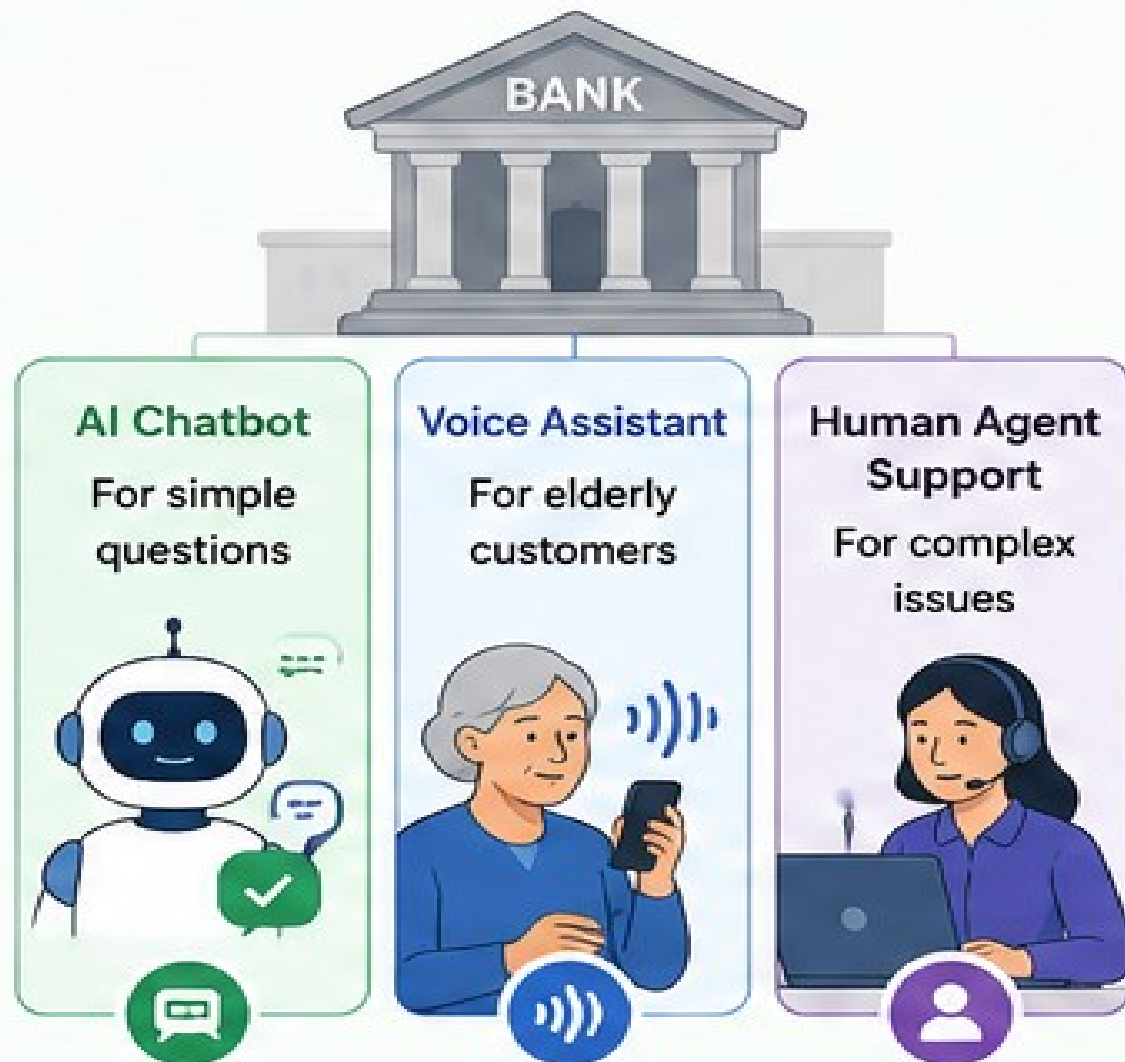


2 Encourages Creative Solutions

Example:

A bank wants an AI customer support system.

The team designs a combination of solutions for different needs.



3 Reduces Risk of Product Failure

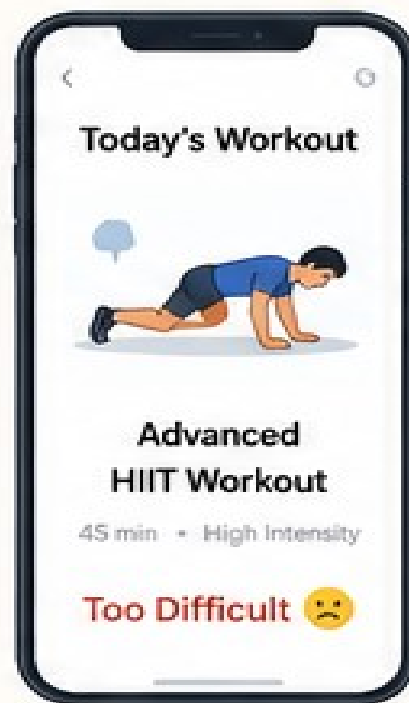
Example:

A startup develops an AI-powered fitness app.

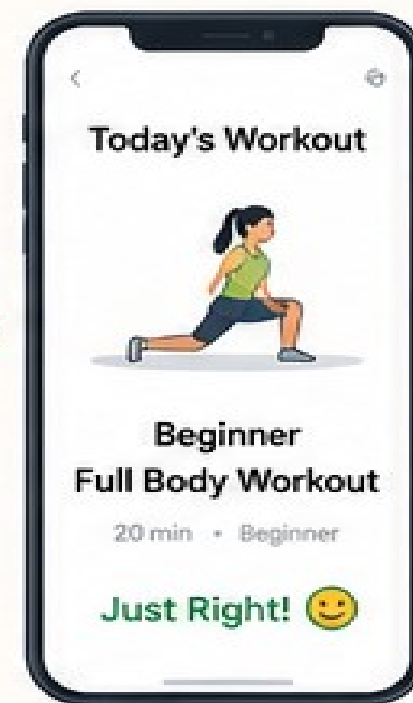
Users find workouts too difficult for beginners in prototype testing.

The AI is updated to personalize exercises before launch.

Before Testing



After Improvement



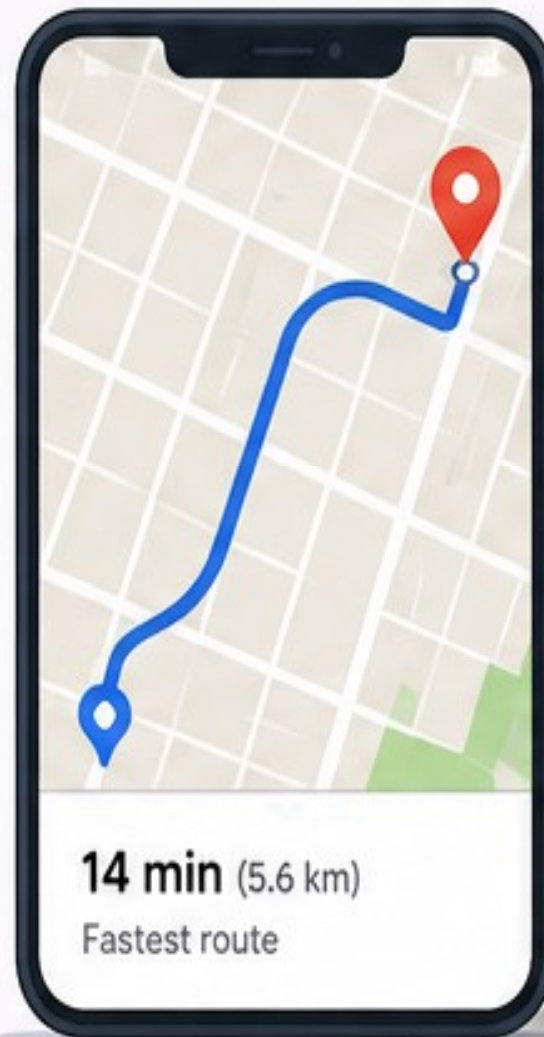
Personalized workouts based on fitness level

4 Improves User Experience (UX)

Example:

Google Maps uses AI to recommend the fastest route.

It explains the reasons clearly so users can trust and make informed decisions.



 5 minutes faster

 Avoids heavy traffic

 Road construction ahead



5 Helps Build Better AI Products

Example:

Netflix uses AI to recommend movies and TV shows.

It learns user preferences from viewing habits and feedback to suggest content users love.



Design Thinking Process

Empathize



Learn About Your Audience

Define



Construct Point of View Based on User Needs

Ideate



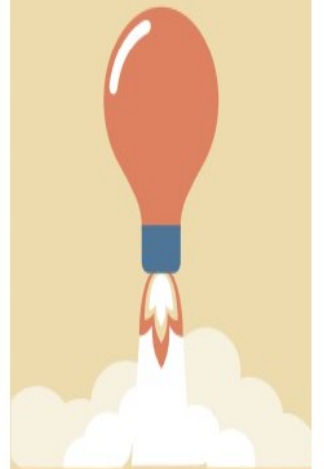
Brainstorm and Come up with Creative Solutions

Prototype



Build Representation of Your Ideas

Test



Test Your Ideas

1. Empathize (Learn About Your Audience)



- Understand the users, their needs, emotions, and challenges.
- Gather information through interviews, surveys, and observations.
- Identify users' goals, pain points, and expectations.
- Build AI solutions that focus on real human needs.

2. Define (Construct Point of View Based on User Needs)



- Analyze the information collected during the empathize stage.
- Clearly identify and define the core problem to be solved.
- Create user personas and problem statements.
- Ensure the AI system addresses the right user issue.

3. Ideate (Brainstorm and Come Up with Creative Solutions)



- Generate multiple ideas and possible AI-based solutions.
- Encourage creativity without judging ideas initially.
- Explore different algorithms, features, and user interactions.
- Select the most practical and user-friendly solution.

4. Prototype (Build Representation of Your Ideas)



- Develop a simple model or prototype of the AI system.
- Create wireframes, mockups, or a minimum viable product (MVP).
- Demonstrate how the AI solution will work.
- Identify design flaws before full-scale development.

5. Test (Test Your Ideas)



- Test the prototype with real users.
- Collect feedback on usability, accuracy, fairness, and satisfaction.
- Identify errors and areas for improvement.
- Refine and improve the AI system based on user feedback.

1 EMPATHIZE

Learn about
your audience



2 DEFINE

Construct point of view
based on user needs



3 IDEATE

Brainstorm and come up
with creative solutions



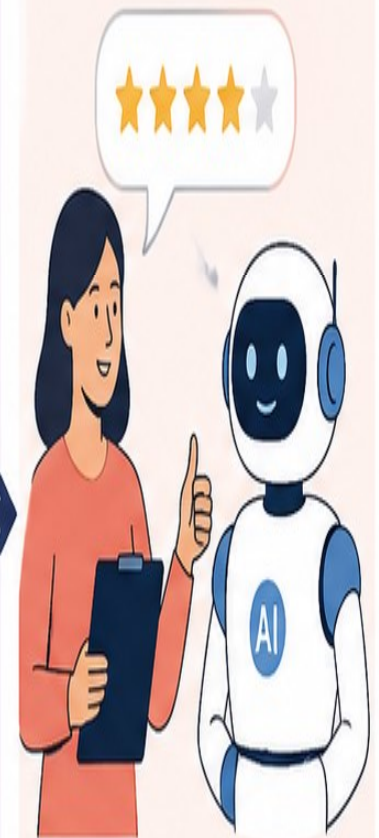
4 PROTOTYPE

Build representation
of your ideas



5 TEST

Test your ideas



Empathy Mapping



- Empathy Mapping is a technique used to **understand the thoughts, emotions, behaviors, and experiences of users.**
- An empathy map usually contains four sections:
- **Says** – What the user says during interviews or conversations.
- **Thinks** – What the user is thinking but may not express directly.
- **Does** – Actions and behaviors of the user.
- **Feels** – Emotions and feelings experienced by the user.

Model

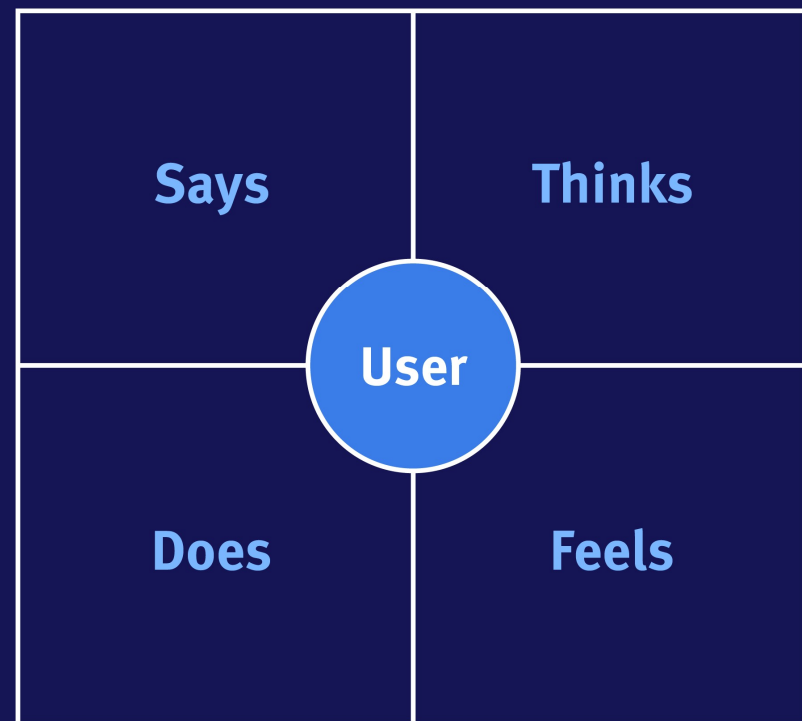


Empathy Mapping:

The First Step in Design Thinking

nngroup.com

NN/g



Ex : AI-powered "Cricket Buddy Finder"



- **It uses location, skill level, availability, and preferences to connect players and organize games.**



Says

- "I miss playing cricket; it's my favourite sport."
- "Finding cricket buddies in Bangalore has been challenging."
- "I wish I could find someone who's as passionate about cricket as I am."

Thinks

- "I hope there's an easy way to connect with cricket enthusiasts nearby."
- "It would be great to organise matches or join existing ones without much hassle."

Feels

- Frustrated about not being able to play cricket regularly since moving to Bangalore.
- Excited about the prospect of finding cricket buddies.
- Anxious about meeting new people but also eager to make friends through

Does

- Searches online for local cricket clubs or groups.
- Occasionally visits local cricket grounds in hopes of finding a game.
- Feels hesitant to approach strangers to organize cricket matches



This empathy map is describing the needs of a user for an AI-based app that could:



- Match players with nearby cricket enthusiasts using AI.
- Recommend local matches and cricket grounds.
- Form balanced teams based on players' skill levels.
- Suggest suitable playing times and venues.
- Organize matches automatically.
- Help users build a cricket-playing social network.

Why use this for AI?



- Usually, AI is built to be "smart" or "fast." But if an AI is smart but feels **creepy** or **annoying**, people won't use it.
- **Example:** If an AI medical app detects a health issue, an empathy map helps developers realize the user feels **scared**. Instead of a cold, robotic message, they program the AI to use a **gentle, supportive tone**.
- In short: It's about making sure the "Brain" (AI) has a "Heart" (Empathy).

Example 1: AI Healthcare Assistant



Persona: John, 65 years old, managing diabetes.

Says

"Did I take my medicine?"

"Remind me about my appointment."

Does

Checks health app daily

Records blood sugar levels

Thinks

"I don't want to miss my medication."

"I hope my health stays stable."

Feels

Worried about health

Happy when reminders help

Example 2: AI Shopping Assistant



Persona: Emma, an online shopper.

Says

Thinks

"Show me cheaper options."

"I want the best deal."

"Compare these products."

"Can I trust these reviews?"

Does

Feels

Reads reviews

Excited about discounts

Uses price comparison

Frustrated by too many choices

Example 3: AI Travel Planner



Persona: Alex, planning a vacation.

Says

"Find the cheapest flight."

"Plan a 3-day itinerary."

Thinks

"I hope I don't miss anything."

"I want to save time."

Does

Searches hotels and flights

Checks weather forecasts

Feels

Excited about the trip

Stressed about planning

User Persona Development



- An AI User Persona is a detailed, data-driven representation of a target audience created using artificial intelligence.
- To create an AI User Persona, you typically start by gathering data from various sources, like:
- Surveys.
- Customer Interviews.
- Customer support.
- Feedback Forms.
- Social media.

How to Build an AI User Persona

1



Step 1:
User Data Collection

2



Step 2:
Define Key Persona Traits

3



Step 3:
Personalize the Persona

Example: Creating an AI User Persona for an Online Food Delivery App



- Imagine a company like **Zomato** or **Swiggy** wants to understand its customers better and improve its app experience.
- They use AI to collect and analyze customer data from different sources.

1. SURVEYS

Collect data from surveys sent to users.



Example Questions:

- ✓ How often do you order food online?
- ✓ What type of food do you prefer?
- ✓ What problems do you face while ordering?

User Responses:

- 💬 "I order food 4 times a week."
- 💬 "I prefer healthy food options."
- 💬 "Delivery time is my biggest concern."



AI analyzes thousands of survey responses and finds patterns.

2. CUSTOMER INTERVIEWS

Direct conversations with customers.



Example Conversation:

Customer:

"I work from 9 AM to 7 PM, so I usually order dinner after work. I don't have time to compare many restaurants."



AI Identifies:

- User is a working professional.
- User values speed and convenience.
- User needs quick restaurant recommendations.

3. CUSTOMER SUPPORT DATA

Analyze complaints and support chats.



My food arrived late.

The app suggested restaurants too far away.

I want more vegetarian options.



AI Identifies Common Problems:

- Delivery speed is important.
- Location-based suggestions need improvement.
- Food preferences matter.

4. FEEDBACK FORMS

Analyze ratings and feedback after orders.

How was your experience?



2 / 5

"Food was good, but delivery took 60 minutes."



AI Learns:

- Customer likes food quality.
- Customer dislikes long waiting times.

5. SOCIAL MEDIA DATA

Analyze social media comments, posts, and reviews.



Twitter/X:

“Why does the app show expensive restaurants first?”



Instagram:

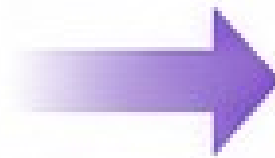
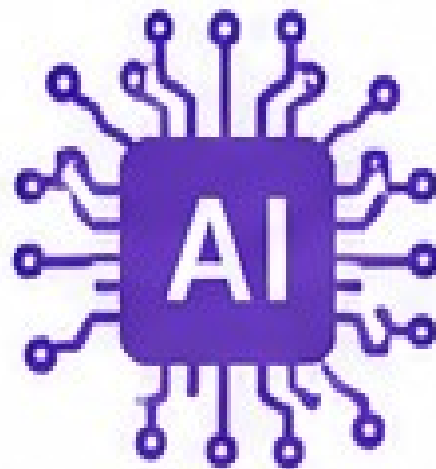
“Love the weekend discounts!”



AI Discovers:

- Users care about price.
- Discounts influence purchase decisions.

AI analyzes all the data and creates
a fictional but data-based customer profile.



6. AI CREATES USER PERSONA

Name: Priya Sharma

Age: 28 years old

Occupation: Software Engineer

Location: Bangalore, India

Lifestyle:

- Works long hours.
- Orders food 4–5 times per week.
- Uses mobile apps frequently.

Goals

- ✓ Get food quickly after work
- ✓ Find healthy meal options
- ✓ Save money using offers

Problems (Pain Points)

- ✗ Long delivery times
- ✗ Too many restaurant choices
- ✗ Expensive meals

Behavior

- Orders mainly between 8 PM–10 PM.
- Prefers restaurants within 5 km.
- Checks ratings before ordering.

Preferred Features

- AI food recommendations
- Fast delivery option
- Personalized discounts
- Healthy food filters

How the Company Uses This AI Persona

Takes too long to
choose food



AI recommends meals
based on past orders

Wants faster
delivery



Shows "30-minute
delivery" restaurants

Likes discounts



Sends personalized
coupons

Prefers healthy
food



Adds health-based
recommendations

Why is User Persona Important in AI Product Design



- **Understands Target Users** – Identifies who will use the AI product.
- **Identifies User Needs** – Helps design features based on user requirements.
- **Finds Pain Points** – Reveals problems that AI should solve.
- **Improves User Experience (UX)** – Makes the product easy and comfortable to use.
- **Supports Personalization** – AI provides recommendations based on user preferences.
- **Saves Time and Cost** – Reduces unnecessary features and redesign.
- **Increases User Satisfaction** – Creates AI products that better meet user expectations.

Problem Statement Writing



- A **problem statement in HCAI** describes a real human problem that an AI system should solve while keeping **people's needs, experiences, and limitations at the center.**

A good HCAI problem statement usually answers:



- **Who is facing the problem? (Target users)**
- **What problem are they facing?**
- **Why is this problem important?**
- **What is the impact of the problem?**
- **How can AI help improve the situation?**



Components of a Good HCAI Problem Statement

1. Target User (Who?)



- **Examples:**
- Students
- Teachers
- Patients
- Customers
- Employees

2. Problem (What?)



- What difficulty are they experiencing?
- **Example:**
- Students struggle to manage study schedules.

3. Impact (Why?)



- What difficulty are they experiencing?
- **Example:**
- Students struggle to manage study schedules.

4. AI Solution (How?)



- How can AI help?
- **Example:**
- AI can provide personalized study plans and reminders.

5. Human Values



- How will the solution remain human-centered?
- **Examples:**
- Privacy
- Trust
- Fairness
- Transparency
- User control

Example 1: Healthcare AI System



- **Poor Problem Statement:**
- "Create an AI system that reminds people to take medicines."
- **Why it is poor:**
- Only describes a feature.
- Does not explain who needs it.
- Does not mention the user's difficulties.

HCAI Problem Statement:



- "Many elderly patients forget to take medicines at the correct time, which can affect their health. They need a simple AI-based reminder system that provides personalized alerts and allows caregivers to support them while protecting their privacy."
- **Why it is good:**
- Identifies users → Elderly patients and caregivers.
- Explains the problem → Forgetting medicines.
- Provides AI solution → Personalized reminders.
- Includes human concerns → Privacy and support.

Example 2: Education AI System



- **Poor Problem Statement:**
- "Create an AI system that teaches students."
- **Why it is poor:**
- Focuses only on AI technology.
- Does not explain the actual problem.
- Does not consider students' needs.

HCAI Problem Statement:



- Many students struggle to understand lessons because they have different learning speeds and abilities. They need an AI learning assistant that provides personalized explanations and supports teachers while ensuring students' privacy and control.“
- **Why it is good:**
- Identifies the users → Students and teachers.
- Defines the real problem → Difficulty in understanding lessons.
- Explains the impact → Learning difficulties.
- Shows AI's role → Personalized learning support.
- Considers human values → Privacy and control.

Customer Journey Mapping



- Customer Journey Mapping is a method used to **visualize the entire experience of a user while interacting with a product or service.**
- It shows the steps a user takes while using a system, such as:
 - Discovering the product
 - Exploring features
 - Using the product
 - Getting support
 - Providing feedback

Stages of a Customer Journey Map



Component	Description
Customer (Persona)	Who is using the product or service?
Stages	Steps the customer goes through
Actions	What the customer does at each stage
Thoughts & Feelings	What the customer thinks or feels
Pain Points	Problems faced by the customer
Opportunities	Ways to improve the experience using AI or design

Customer Persona

Name: Rahul

Age: 24

Occupation: College Student

Stage	Customer Action	Feeling	Pain Point	AI Opportunity
Awareness	Sees food delivery ad	Curious	Too many apps	AI recommends popular app
Consideration	Compares restaurants	Confused	Too many choices	AI suggests restaurants based on preferences
Purchase	Places an order	Happy	Payment issues	AI detects payment errors
Delivery	Tracks the order	Excited	Delivery delay	AI provides accurate delivery time
Support	Contacts support	Frustrated	Slow response	AI chatbot gives instant answers
Loyalty	Orders again	Satisfied	Wants discounts	AI offers personalized coupons

Why is Customer Journey Important in HCAI?



- Understands customer needs.
- Identifies pain points (problems).
- Helps design AI solutions that are useful and user-friendly.
- Ensures AI supports people at every stage.

CUSTOMER JOURNEY MAPPING IN AI PRODUCT DESIGN



User Persona

Rahul, 20 years old
Diploma Student

Example: AI Study Assistant (Mobile App)

Goal: Learn better, complete assignments on time, score well

STAGES	1. AWARENESS	2. CONSIDERATION	3. ONBOARDING	4. USAGE	5. SUPPORT	6. LOYALTY
Customer Actions 	Sees ad on social media about the AI Study Assistant. 	Reads reviews, compares features with other study apps. 	Downloads the app, creates account, sets learning goals. 	Uses AI features like notes summary, doubt solve, quiz, reminders. 	Faces an issue and contacts support for help. 	Continues using the app, gives good rating and recommends to friends.
Customer Thoughts 	This app looks useful. I should check it.	Is this better than other apps? Will it really help me?	Hope it is easy to use and understand.	This AI feature is helping me save time!	I hope I get quick and accurate support.	This app is really helpful. I will suggest to others.
Customer Feelings	Curious	Unsure	Optimistic	Satisfied	Frustrated	Happy & Loyal
 Pain Points	Not sure if the app is trustworthy.	Too many apps with similar features.	Sign up process is long.	Some features are hard to understand.	Waiting time for live support is high.	Wants more personalized recommendations.
AI Opportunities & Solutions 	AI-powered targeted ads and clear value communication. 	AI compares user needs and shows best app features. 	AI-based quick sign-up and personalized learning setup. 	AI provides personalized content, smart reminders, progress tracking. 	AI chatbot for instant answers and issue resolution. 	AI suggests personalized study plans and rewards for continued use.
OVERALL JOURNEY EMOTION						

Customer Journey Mapping helps us understand user needs, pain points and how AI can create a better, smarter and more human-centered experience.



UX Principles for AI Systems

1. Human-Centered Design



- Design the AI system according to users' needs and goals.
- AI should assist users, not replace them completely.
- Focus on solving real user problems.
- **Example:** An AI study assistant helps students learn but does not replace teachers.

2. Transparency



- The AI should explain how or why it made a decision or recommendation.
 - Users should understand the AI's actions.
 - This builds trust and confidence.
-
- **Example:** Netflix explains, "*Recommended because you watched action movies.*"

3. User Control and Feedback



- Users should always have control over AI decisions.
 - Allow users to accept, reject, or modify AI suggestions.
 - Collect user feedback to improve the AI system.
-
- **Example:** Google Maps suggests a route, but users can choose a different one.

4. Trust, Privacy, and Security



- Protect users' personal data.
 - Use data responsibly and securely.
 - Clearly inform users how their data is collected and used.
-
- **Example:** A healthcare AI app keeps patient records secure and confidential.

5. Fairness and Bias Reduction



- AI should treat all users fairly.
- Avoid discrimination based on gender, age, race, or other personal characteristics.
- Train AI using diverse and balanced data.
- **Example:** An AI recruitment system should evaluate candidates based on skills, not personal background.

6. Simplicity and Usability



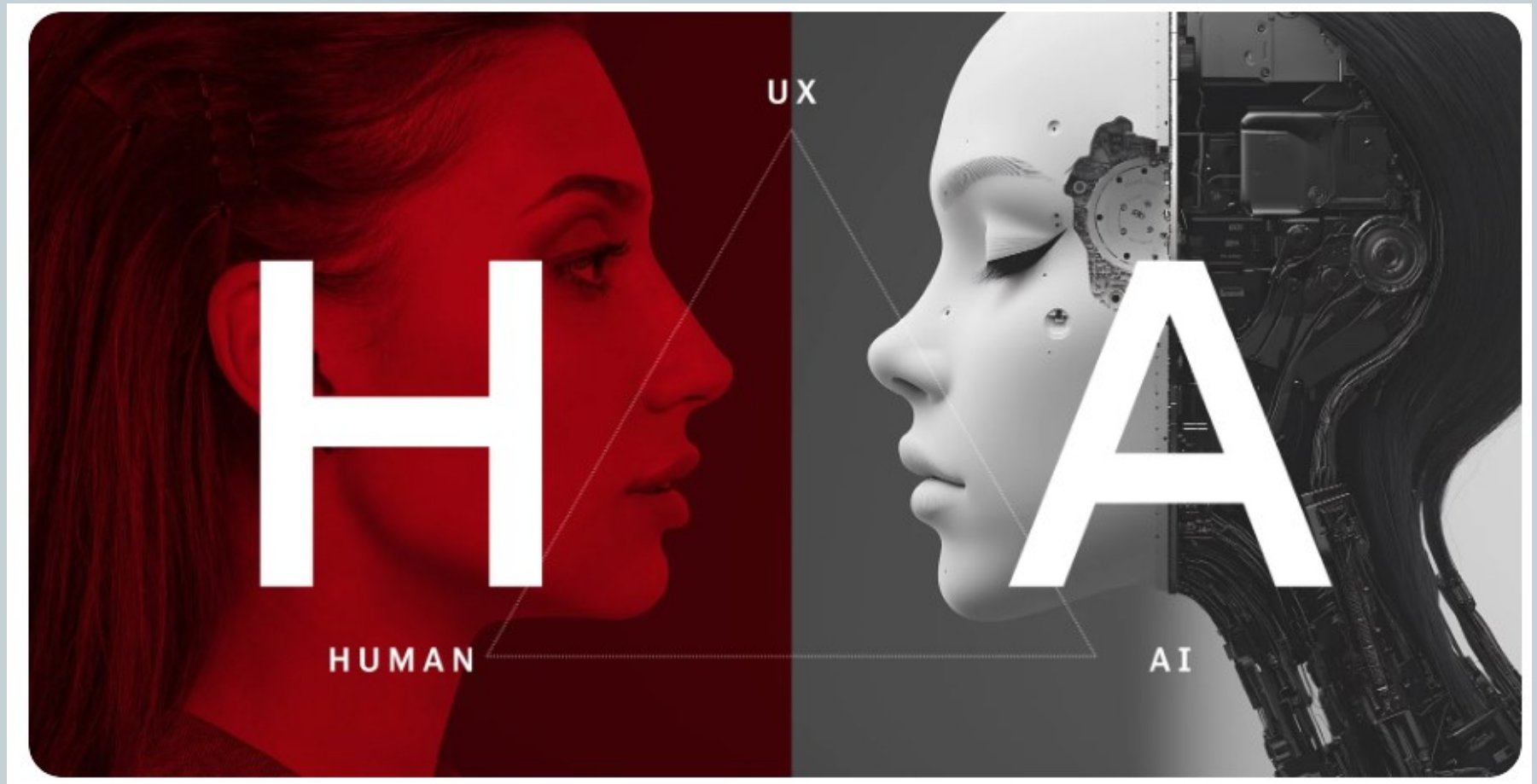
- The interface should be simple, clear, and easy to use.
- Reduce complexity and avoid confusing interactions.
- Make AI accessible to users with different skill levels.
- **Example:** A voice assistant allows users to perform tasks using simple voice commands.

7. Continuous Learning and Improvement



- AI should improve over time using user feedback and new data.
- Regularly update the system to increase accuracy and user satisfaction.
- Fix errors and adapt to changing user needs.
- **Example:** A music streaming app improves song recommendations based on listening history.

Human-AI Interaction Design



Human-AI Interaction Design (HAX)



- Human-AI Interaction Design focuses on how **people interact with artificial intelligence systems**. The goal is to create systems where humans and AI work together effectively.
- In good Human-AI interaction:
- AI assists humans instead of replacing them.
- Users understand how the AI system works.
- Users can easily interact with the AI system.
- Humans can override AI decisions if necessary.

Objectives



- Improve user experience.
 - Build trust in AI.
 - Make AI easy to understand.
 - Help humans make better decisions.
-
- **Example:** Google Maps suggests the fastest route, but the driver decides whether to follow i

Components of Human-AI Interaction



- 1. Human (User) : The person who uses the AI system.
 - Example: Student, Doctor, Customer, Teacher
- 2. AI System : The intelligent software that performs tasks or provides recommendations.
 - Example: ChatGPT, Siri, Google Assistant, Netflix Recommendation System
- 3. Interaction : The communication between the human and the AI.
 - Example: Asking questions, Giving voice commands, Clicking recommendations, Receiving suggestions

Characteristics of Good Human-AI Interaction



- **1. User-Friendly**
- **2. Transparency**
- **3. Trustworthy**
- **4. Human Control**
- **5. Privacy**
- **6. Fairness**
- **7. Continuous Learning**

Levels of Human-AI Interaction



- **1. Human Controls AI**
- The human gives commands, and AI follows them.
- **Example:**
Voice assistants like Alexa.



- **2. Human and AI Collaborate**

- Both work together to complete a task.

- **Example:**

AI helps doctors detect diseases, while doctors make the final decision.



- **3. AI Automates Tasks**

- AI performs repetitive tasks automatically, but humans supervise.

- **Example:**

AI Traffic Signal System

- **AI automates:** Adjusting traffic signal timings based on traffic density.
- **Human supervises:** Traffic police monitor the system and take control during emergencies or special events.

Real-Life Examples of HAX

Field	Human	AI System	Interaction
Healthcare	Doctor	AI Diagnosis Tool	AI suggests diseases; doctor confirms
Education	Student	AI Tutor	Student asks questions; AI explains
Banking	Customer	AI Chatbot	Customer asks balance; AI replies
Shopping	Buyer	Recommendation System	AI suggests products
Travel	Tourist	AI Travel Planner	AI recommends hotels and routes

Advantages of HAX



- Saves time.
- Improves productivity.
- Reduces errors.
- Provides personalized recommendations.
- Improves decision-making.

Challenges of HAX



- Privacy concerns.
- AI bias.
- Lack of transparency.
- Incorrect AI responses.
- Overdependence on AI.
- Security risks.

Bias in AI systems



- Bias in AI occurs when an AI system produces **unfair, inaccurate, or discriminatory results**. This usually happens because the training data contains biases or because the model is designed improperly.
- For example, if an AI system is trained mostly on data from one group of people, it may perform poorly for other groups.

Why Does AI Bias Occur?/ Types of AI Bias



- **1. Data Bias**

- Data bias occurs when the training data is incomplete, unbalanced, or does not represent all groups equally.

- **Example**

- A facial recognition system is trained mostly on images of adults, so it performs poorly for children.

Why Does AI Bias Occur?/ Types of AI Bias



- **2. Selection Bias**

Selection bias occurs when the data collected represents only a specific group of people instead of the entire population.

- **Example**

- A fitness app collects data only from young users and uses it to recommend exercise plans for everyone.

Why Does AI Bias Occur?/ Types of AI Bias



- **3. Algorithm Bias**

- Algorithm bias occurs when the AI algorithm or model produces unfair results because of the way it is designed or optimized.

- **Example**

- A job recommendation system always ranks candidates from certain universities higher than equally qualified candidates from other colleges.

Why Does AI Bias Occur?/ Types of AI Bias



- **4. Human Bias**

- Human bias occurs when developers, data labelers, or decision-makers unintentionally introduce their personal beliefs or preferences into the AI system.

- **Example**

- People labeling images assume that "engineer" is always a man and "nurse" is always a woman.

Why Does AI Bias Occur?/ Types of AI Bias



- **5. Confirmation Bias**

- Confirmation bias occurs when AI repeatedly recommends information similar to a user's previous choices, ignoring different viewpoints.

- **Example**

- A social media platform keeps showing news articles that match a user's opinions.

Example



- An AI loan system rejects applicants from certain areas because historical data shows fewer approved loans there.
- An AI translator assumes "doctor" is male and "nurse" is female based on patterns in its training data.
- The AI was trained mainly using images of light skin, so it is less accurate for patients with darker skin.

Effects of AI Bias



- Unfair decisions.
- Discrimination against certain groups.
- Loss of user trust.
- Reduced accuracy for some users.
- Ethical and legal issues.

How to Reduce AI Bias



- **1. Use Diverse and Balanced Data**
 - Train AI using data that represents different genders, ages, cultures, and backgrounds.
- **2. Test AI Regularly**
 - Evaluate AI with different user groups to identify and correct bias.
- **3. Increase Transparency**
 - Explain how AI makes decisions so users can understand and question the results.
- **4. Human Oversight**
 - Humans should review important AI decisions instead of relying completely on AI.
- **5. Update the AI System**
 - Continuously improve the AI using new, fair, and unbiased data.

Explainable AI (XAI)



Explainable AI



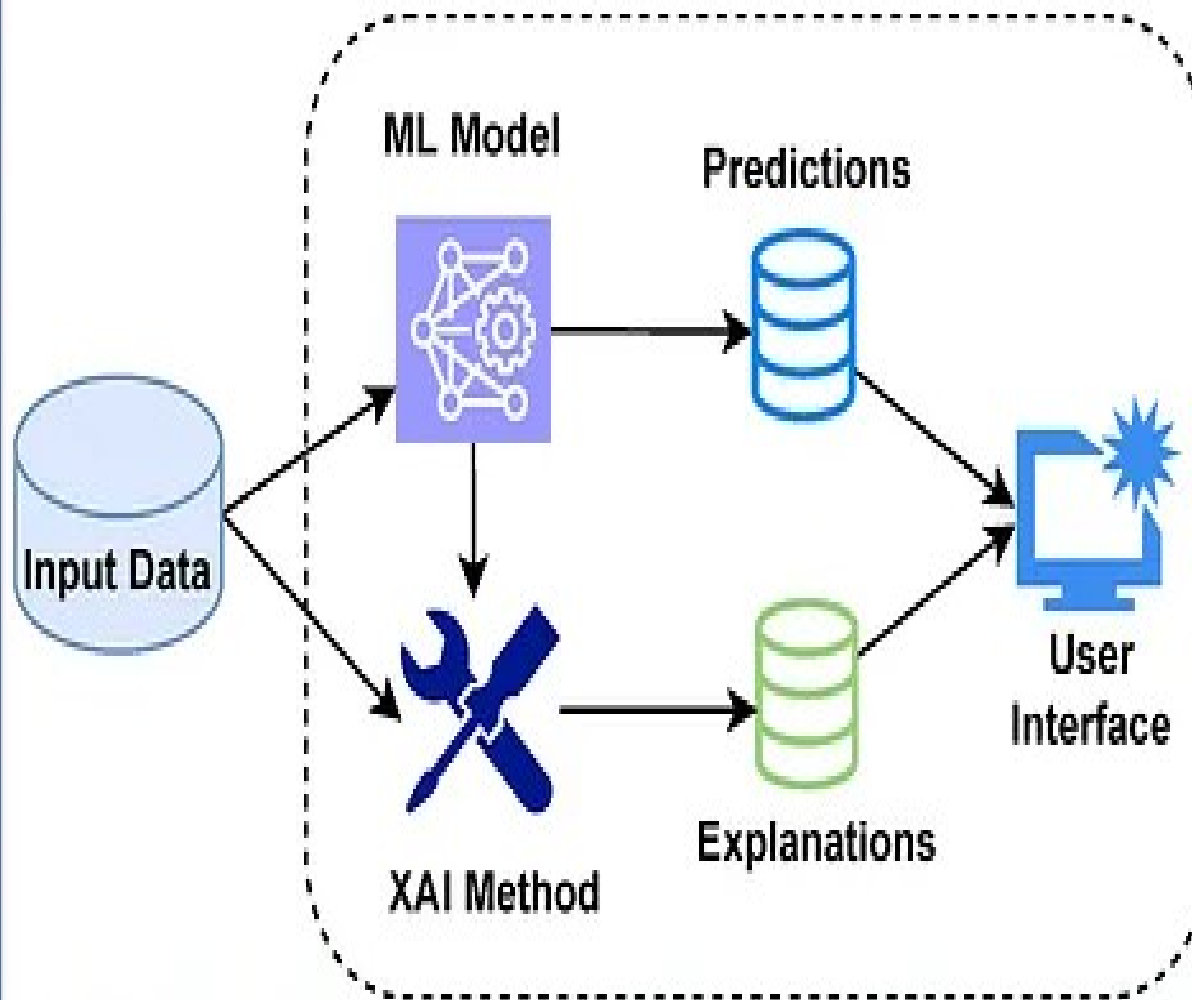
- It is about making AI more “explainable” so that we can understand why a certain decision was made, how it was made, and how we can trust it.
- Explainable AI answers questions like:
- Why did the AI make this decision?
- What factors influenced the result?
- How reliable is the prediction?

How Explainable AI Works



- User gives input.
- AI analyzes the data.
- AI makes a prediction or decision.
- AI provides the result **with an explanation**.
- User understands and decides whether to accept the result.

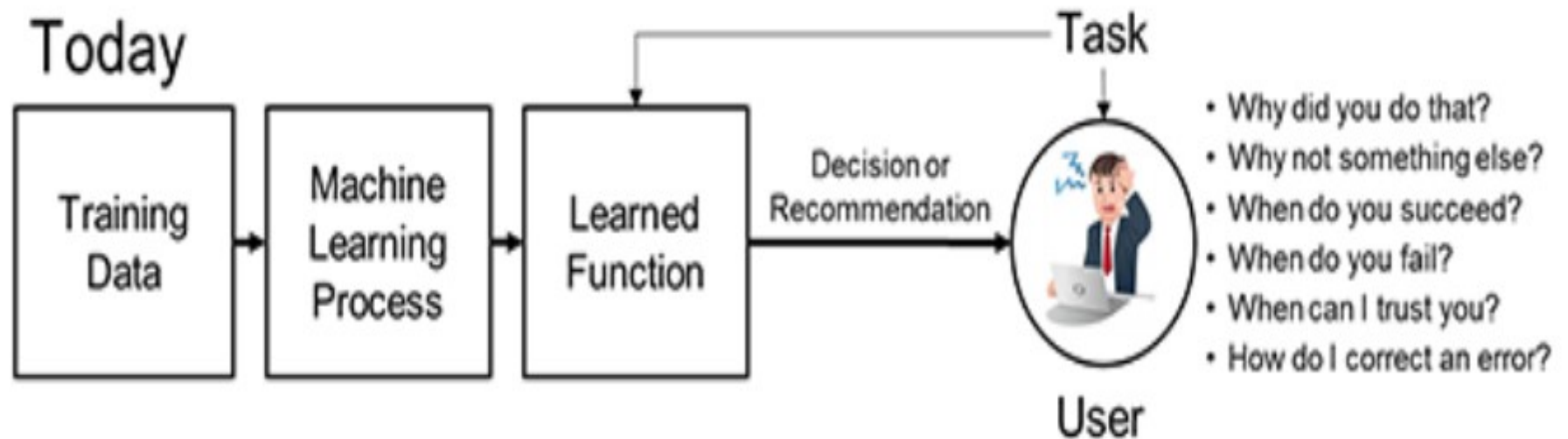
Explainable AI (XAI)



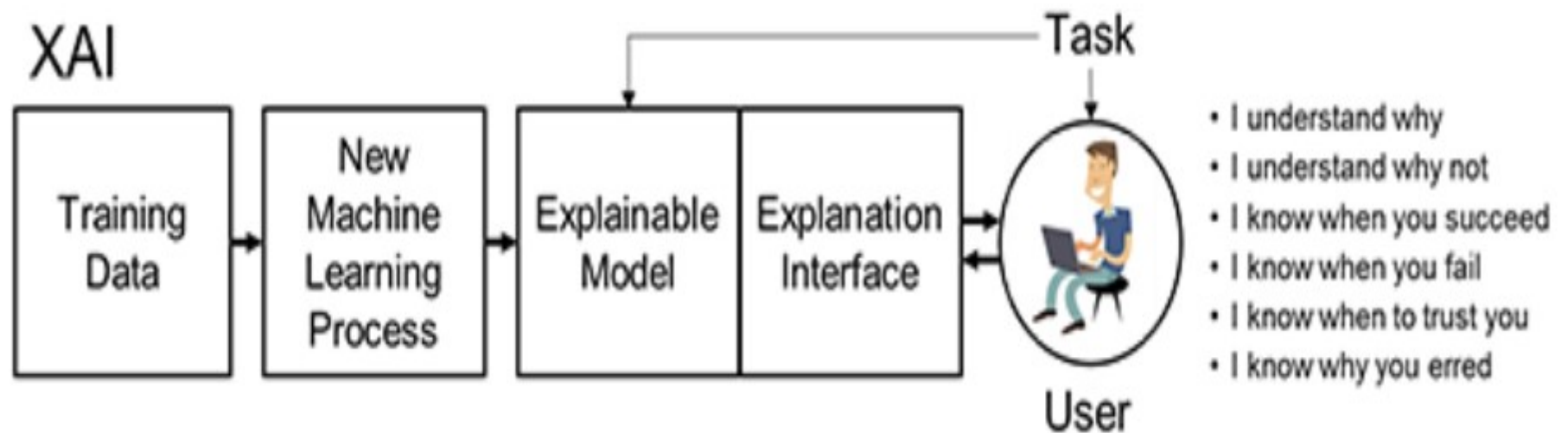
Stakeholders



Today



XAI



Real-Time Examples of Explainable AI



- **1. Banking (Loan Approval)**
- **Without XAI:**
- Loan Rejected.
- **With XAI:**
- Loan rejected because your credit score is 580 (minimum required is 700) and your monthly income does not meet the eligibility criteria.

Real-Time Examples of Explainable AI



- **2. Healthcare**

- **Without XAI:**

- Disease detected.

- **With XAI:**

- The AI predicts pneumonia because the chest X-ray shows abnormal lung patterns similar to previous pneumonia cases.
- The doctor understands the reason before making the final diagnosis.

Real-Time Examples of Explainable AI



- **3. Education**

- **Without XAI:**

- You scored 65%.

- **With XAI:**

- Your score is 65% because questions related to programming loops and functions were answered incorrectly.
- Students know where they need improvement.

Real-Time Examples of Explainable AI



- **4. Navigation App**
- **Without XAI:**
 - Take Route A.
- **With XAI:**
 - Route A is recommended because it is 10 minutes faster and has less traffic than Route B.

Characteristics of Explainable AI



- **1. Transparency**
- **2. Interpretability**
- **3. Accountability**
- **4. Trustworthiness**
- **5. Human Control**

Explainable AI vs Traditional AI

Traditional AI

Gives only the result

Decision process is hidden

Hard to trust

Difficult to identify errors

Users cannot understand why

Explainable AI (XAI)

Gives the result with a clear explanation

Decision process is transparent

Easier to trust

Easier to detect errors and bias

Users understand the reason behind decisions

3–4 Mark Questions



- Define Design Thinking. Why Design Thinking is important?
- Discuss Empathy Mapping and explain the four components of an Empathy Map.
- What is a User Persona? Why are User Personas important in product design?
- What is Customer Journey Mapping and why is it important in AI product Design?
- Define Human-AI Interaction Design (HAX) and what are the objectives of it?
- Discuss components and characteristics of HAX.
- Discuss advantages and challenges of HAX.
- Short Note on : AI Bias System
- Short Note on : Explainable AI (XAI)?
- Differentiate Traditional AI and Explainable AI.

7 Mark Questions



- Explain the **Design Thinking process** with its stages.
- Draw Empathy map for (a) AI travel planner (2) AI shopping assistant (3) AI tutor (4) AI health assistant.
- Describe the **process of User Persona Development** with an example.
- What are the different components for a good HCAI **Problem Statement Writing** in Design Thinking.
- Draw a customer journey map for (a) AI travel planner (2) AI shopping assistant (3) AI tutor (4) AI health assistant.
- Discuss the **UX principles for AI systems**.
- Explain **Human-AI Interaction Design** with its different levels and real-world examples.
- Explain the concept of **Bias in AI systems** and its types in detail. / Why does AI Bias Occurs?
- What are the effects of AI Bias and how to reduce its effects?
- Describe **Explainable AI** and its real time examples.